Chapter-9

Inverse Circular Functions

- **Inverse sine function**

→ If $x = \sin y$ be a function defined on $y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ then inverse sine function is denoted by $\sin^{-1}$ or arcsin and is defined by $y = \sin^{-1} x \Leftrightarrow x = \sin y$ provided that $x \in [-1, 1]$ & $y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

Similarly
- inverse cosine function

$$y = \cos^{-1} x \Leftrightarrow x = \cos y$$

Domain - $[-1, 1]$
Range - $[0, \pi]$

- **Inverse tangent function**

$$y = \tan^{-1} x \Leftrightarrow x = \tan y$$

Domain - $[-\infty, \infty)$
Range - $(-\pi/2, \pi/2)$

I. For appropriate θ,
i) $\sin^{-1} \sin\theta = \theta$
ii) $\cos^{-1} \cos\theta = \theta$
iii) $\tan^{-1} \tan\theta = \theta$ etc.

i) Proof :-

Let $\sin\theta = x$

By definition, $\theta = \sin^{-1}x$

$\theta = \sin^{-1}\sin\theta$

$\therefore \sin^{-1}\sin\theta = \theta$

proved.

ii) Proof

let $\cos\theta = x$

By definition, $\theta = \cos^{-1}x$

$\theta = \cos^{-1}\cos\theta$

$\therefore \cos^{-1}\cos\theta = \theta$ pd.

II) For appropriate value of x.

i) $\sin\sin^{-1}x = x$

let $\sin^{-1}x = x$

By definition $x = \sin x$.

$x = \sin\sin^{-1}x$

$\therefore \sin\sin^{-1}x = x$ p.

III)

i) $\sin^{-1}x = \operatorname{cosec}^{-1}\frac{1}{x}$

let $\sin^{-1}x = \theta$

$\sin\theta = x$

$\frac{1}{\operatorname{cosec}\theta} = x \Rightarrow \operatorname{cosec}\theta = \frac{1}{x}$

$\theta = \operatorname{cosec}^{-1}\frac{1}{x}$

$\therefore \sin^{-1}x = \operatorname{cosec}^{-1}\frac{1}{x}$ pd

IV. ~~Sr.~~

i) $\sin^{-1} x = \cos^{-1}\sqrt{1-x^2}$

ii) $\cos^{-1} x = \sin^{-1}\sqrt{1-x^2}$

i) Proof

let $\sin^{-1} x = \theta$.

$\sin\theta = x$

$L.H.S = \sin^{-1} x = \theta$

$R.H.S = \cos^{-1}\sqrt{1-x^2} = \cos^{-1}\sqrt{1-\sin^2\theta}$

$= \cos^{-1}\cos\theta$

$= \theta$

$\therefore \sin^{-1} x = \cos^{-1}\sqrt{1-x^2} = \theta$

iii) ~~Proof~~ $\sin^{-1} x = \tan^{-1}\dfrac{x}{\sqrt{1-x^2}}$

iv) $\sin^{-1} x = \cot^{-1}\dfrac{\sqrt{1-x^2}}{x}$

iii) Proof

Let, $\sin^{-1} x = \theta$

$x = \sin\theta$

$R.H.S = \tan^{-1}\dfrac{x}{\sqrt{1-x^2}}$

$= \tan^{-1}\dfrac{\sin\theta}{\sqrt{1-\sin^2\theta}}$

$= \tan^{-1}\dfrac{\sin\theta}{\cos\theta}$

$= \tan^{-1}\tan\theta$

$= \theta$

$= \cancel{\sin\theta}\ \sin^{-1} x = L.H.S$

V)

i) $\sin^{-1}(-x) = -\sin^{-1}x$

ii) $\tan^{-1}(-x) = -\tan^{-1}x$

iii) $\cos^{-1}(-x) = \pi - \cos^{-1}x$

i) Proof

let $\sin^{-1}(-x) = \theta$.

$-x = \sin\theta$

$x = -\sin\theta$.

R.H.S $= -\sin^{-1}x$

$= -\sin^{-1}(-\sin\theta)$

$= -\sin^{-1}\sin(-\theta)$

$= -(-\theta)$

$= \theta$

~~iii) Proof~~

~~let $\cos^{-1}(-x) = \theta$.~~

~~$-x = \cos\theta$.~~

~~$x = -\cos\theta$.~~

~~R.H.S $= \pi - \cos^{-1}x$~~

~~$= \pi - \cos^{-1}(\cos(\theta))$~~

~~$= \pi - \cos^{-1}(\cos\theta($~~

~~$= \pi - \theta$~~

iii) Let $\cos^{-1}x = \theta$

$\cos\theta = x$

$-\cos\theta = -x$

$\cos(\pi - \theta) = -x$

$L.H.S = \cos^{-1}\cos(\pi-\theta)$
$= \pi-\theta$

$R.H.S = \pi - \cos^{-1}x = \pi-\theta$
$L.H.S = R.H.S$

VI.

i) $\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$

ii) $\operatorname{cosec}^{-1}x + \sec^{-1}x = \frac{\pi}{2}$

iii) $\tan^{-1}x + \cot^{-1}x = \frac{\pi}{2}$

i) Proof
$\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$

let $\sin^{-1}x = \theta$
$\sin\theta = x$
$\Rightarrow \cos\left(\frac{\pi}{2}-\theta\right) = x$

$\Rightarrow \cos^{-1}x = \frac{\pi}{2} - \theta$

$\Rightarrow \cos^{-1}x = \frac{\pi}{2} - \sin^{-1}x$

$\Rightarrow \cos^{-1}x + \sin^{-1}x = \frac{\pi}{2}$

$\therefore \sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$

Date
Page

Exercise - 9

① Evaluate the following

a) $\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$

→ let $\sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = \theta$

$\sin\theta = \frac{1}{\sqrt{2}}$

$\sin\theta = \sin\frac{\pi}{4}$

$\therefore \theta = \frac{\pi}{4}$ //.

b) $\operatorname{cosec}^{-1}(2)$

→ let $\operatorname{cosec}^{-1}(2) = \theta$

$\operatorname{cosec}\theta = 2$

$\frac{1}{\sin\theta} = 2$

$\sin\theta = \frac{1}{2}$

$\sin\theta = \sin 30^\circ$

$\therefore \theta = 30^\circ = \frac{\pi}{6}$ //.

c) $\cot^{-1}(-\sqrt{3})$

let $\cot^{-1}(-\sqrt{3}) = \theta$

$\cot\theta = -\sqrt{3}$

$\frac{1}{\tan\theta} = -\sqrt{3}$

or, $\tan\theta = \frac{-1}{\sqrt{3}}$

or, $\tan\theta = \tan(180-30)$

or, $\tan\theta = \tan 150^\circ = \tan\frac{5\pi}{6}$

$\therefore \theta = 150^\circ = \frac{5\pi}{6}$

d) Arc $\tan\left(\frac{1}{\sqrt{3}}\right)$

let $\tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \theta$

$\tan\theta = \frac{1}{\sqrt{3}}$

$\tan\theta = \tan 30^\circ$

$\theta = 30^\circ$

e) $\sec^{-1}\left(\frac{2}{\sqrt{3}}\right)$

$\rightarrow$ let $\sec^{-1}\left(\frac{2}{\sqrt{3}}\right) = \theta$

$\sec\theta = \frac{2}{\sqrt{3}}$

$\Rightarrow \frac{1}{\cos\theta} = \frac{2}{\sqrt{3}}$

$\Rightarrow \cos\theta = \frac{\sqrt{3}}{2}$

$\Rightarrow \cos\theta = \cos 30^\circ$

$\therefore \theta = 30^\circ$

2) Evaluate:

a) $\cos\left(\tan^{-1}\frac{3}{4}\right)$

→ let $\tan^{-1}\frac{3}{4} = \theta$

$\tan\theta = \frac{3}{4} = \frac{p}{b}$

$h^2 = p^2 + b^2$

$h = \sqrt{3^2+4^2} = \sqrt{9+16}$

$= \sqrt{25}$

$= 5$

Now,

$\cos \tan^{-1}\frac{3}{4}$

$= \cos\theta = \frac{b}{h} = \frac{4}{5}$

$\therefore \cos\left(\tan^{-1}\frac{3}{4}\right) = \frac{4}{5}$

b) $\sin(\cot^{-1}x)$

let $\cot^{-1}x = \theta$

$\cot\theta = \frac{x}{1} = \frac{b}{p}$

$h^2 = p^2 + b^2$

$h = \sqrt{1^2 + x^2}$

$h = \sqrt{x^2+1}$

Now,

$\sin(\cot^{-1}x) = \sin\theta = \frac{p}{h} = \cancel{\sqrt{x^2+1}} \frac{1}{\sqrt{x^2+1}}$

Date
Page

c) $\cancel{\sin^{-1}\left(\sin \frac{2\pi}{3}\right)}$

$= \cancel{\frac{2\pi}{3}}$ $\quad \because \cancel{\sin^{-1} \sin\theta = \theta}$

d) $\cancel{\cos(2\cot^{-1}x)}$

$\cancel{\rightarrow \text{let } 2\cot^{-1}x = \theta}$

$\cancel{2\cot\theta = x}$

$\cancel{\cot\theta = \frac{x}{2}} = \cancel{\frac{b}{p}}$

$\cancel{h = \sqrt{b^2+p^2}}$ $\quad \cancel{= \sqrt{x^2+2^2}}$

$\cancel{= \sqrt{x^2+4}}$

~~Now~~,

$\cancel{\cos(2\cot^{-1}x)} = \cancel{\cos\theta}$

$\cancel{= \frac{b}{h}} = \cancel{\frac{x}{\sqrt{x^2+4}}}$

d) $\cos(2\cot^{-1}x)$.

let $2\cot^{-1}x = \theta$

$\cot^{-1}x = \frac{\theta}{2}$

$\cot\frac{\theta}{2} = \frac{x}{1} = \frac{b}{p}$

$h = \sqrt{p^2+b^2}$

$= \sqrt{1^2+x^2}$

$= \sqrt{1+x^2}$

$\cos(2\cot^{-1}x) = \cos\theta$

$= \cos 2\cdot\frac{\theta}{2}$

Date
Page

$$= \frac{1-\tan^2\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}$$

$$= \frac{1-\frac{1}{x^2}}{1+\frac{1}{x^2}}$$

$$= \frac{\frac{x^2-1}{\cancel{x^2}}}{\frac{x^2+1}{\cancel{x^2}}}$$

$$= \frac{x^2-1}{x^2+1}$$ #

e) $\sin(2\,\text{arc}\tan x)$

$\rightarrow$ ~~let~~ $\sin(2\tan^{-1}x)$

let $\tan^{-1}x = \theta$

$\tan\theta = \frac{x}{1} = \frac{p}{b}$

$h = \sqrt{p^2+b^2} = \sqrt{x^2+1}$

$\sin(2\tan^{-1}x) = \sin 2\theta$

$= 2\sin\theta \cdot \cos\theta$

$= 2\times\frac{p}{h}\times\frac{b}{h}$

$= 2\times\frac{x}{\sqrt{x^2+1}}\times\frac{1}{\sqrt{x^2+1}}$

$= \frac{2x}{x^2+1}$ #

Date
Page

3) Find the value of the following.

a) $\cos\left[\sin^{-1}\frac{4}{5} + \tan^{-1}\frac{5}{12}\right]$

→ let $\sin^{-1}\frac{4}{5} = A$

$\sin A = \frac{4}{5} = \frac{p}{h}$

$h^2 = p^2 + b^2$

$\Rightarrow 5^2 = 4^2 + b^2$

$\Rightarrow b^2 = 25 - 16$

$\Rightarrow b = 9$

$b = \sqrt{9}$

$\therefore b = 3$

$\cos A = \frac{b}{h} = \frac{3}{5}$

Now,

let $\tan^{-1}\frac{5}{12} = B$

$\tan B = \frac{5}{12} = \frac{p}{b}$

$h = \sqrt{p^2 + b^2}$

$h = \sqrt{5^2 + 12^2} = \sqrt{25 + 144}$

$= \sqrt{169}$

$= 13$

$\cos B = \frac{b}{h} = \frac{12}{13}$ $\sin B = \frac{p}{h} = \frac{5}{13}$

Again,

$\cos\left[\sin^{-1}\frac{4}{5} + \tan^{-1}\frac{5}{12}\right]$

$= \cos[A+B]$

$= \cos A.\cos B - \sin A.\sin B$

$= \frac{3}{5} \times \frac{12}{13} - \cancel{\frac{B}{13}} \frac{4}{\cancel{5}} \times \frac{\cancel{5}}{13}$

$= \frac{36}{65} - \frac{4}{13}$

$= \frac{16}{65}$

b) $\tan[\tan^{-1} x - \tan^{-1} 2y]$

→ Soln,

let $\tan^{-1} x = A$

$\tan A = x$

$\tan A = x = \frac{p}{b}$

$h^2 = p^2 + b^2$

$h = \sqrt{x^2 + 1^2}$

$h = \sqrt{x^2+1}$

~~A+B~~

let $\tan^{-1} 2y = B$

$\tan B = 2y$

Now,

$\tan[\tan^{-1} x - \tan^{-1} 2y] = \tan(A-B)$

$= \frac{\tan A - \tan B}{1 + \tan A.\tan B}$

$= \frac{x - 2y}{1 + x \times 2y}$

$$= \frac{x-2y}{1+2xy}$$

~~$\Leftarrow \sin^{-1}x - \cos^{-1}(-x)$~~

~~let $\sin^{-1}x = A$~~ ~~$\cos^{-1}(-x) = B$~~

~~$\sin A = x$~~ ~~$\cos B = -x$~~

4.(a) $\cot^{-1}3 + \operatorname{cosec}^{-1}\sqrt{5} = \frac{\pi}{4}$

i.e $\tan^{-1}\frac{1}{3} + \operatorname{cosec}^{-1}\sqrt{5} = \frac{\pi}{4}$

let, $\operatorname{cosec}^{-1}\sqrt{5} = \theta$

$\operatorname{cosec}\theta = \sqrt{5} = \frac{h}{p}$

we have, $p^2 + b^2 = h^2$ or $1 + b^2 = 5$

$b^2 = 5-1$

$b^2 = 4$

$b = 2$

Now,

$\tan\theta = \frac{p}{b} = \frac{1}{2}$

$\Rightarrow \theta = \tan^{-1}\frac{1}{2}$

$\therefore \operatorname{cosec}^{-1}\sqrt{5} = \tan^{-1}\frac{1}{2}$

Now,

$\tan^{-1}\frac{1}{3} + \operatorname{cosec}^{-1}\sqrt{5}$ $= \tan^{-1}1$

$= \tan^{-1}\tan\frac{\pi}{4}$

$= \tan^{-1}1/3 + \tan^{-1}1/2$

$= \tan^{-1}\frac{1/3 + 1/2}{1 - 1/3 \cdot 1/2}$ $= \pi/4$ proved

VII

I) i) $\boxed{\tan^{-1}x + \tan^{-1}y = \tan^{-1}\frac{x+y}{1-xy}}$

ii) $\tan^{-1}x - \tan^{-1}y = \tan^{-1}\frac{x-y}{1+xy}$

II)

i) $\sin^{-1}x + \sin^{-1}y = \sin^{-1}(x\sqrt{1-y^2} + y\sqrt{1-x^2})$

ii) $\sin^{-1}x - \sin^{-1}y = \sin^{-1}(x\sqrt{1-y^2} - y\sqrt{1-x^2})$

III)

i) $\cos^{-1}x + \cos^{-1}y = \cos^{-1}(xy - \sqrt{1-x^2}\cdot\sqrt{1-y^2})$

ii) $\cos^{-1}x - \cos^{-1}y = \cos^{-1}(xy + \sqrt{1-x^2}\cdot\sqrt{1-y^2})$

Proof:

i) $\tan^{-1}x + \tan^{-1}y$

let, $\tan^{-1}x = A$ $\quad$ $\tan^{-1}y = B$

$x = \tan A$ $\quad$ $y = \tan B$

$$R.H.S = \tan^{-1}\frac{x+y}{1-xy}$$

$$= \tan^{-1}\left(\frac{\tan A + \tan B}{1 - \tan A\cdot\tan B}\right)$$

$$\cancel{= \tan^{-1}(\tan A + \tan B)}$$

$$= \tan^{-1}(\tan(A+B))$$

$$= A+B$$

$$= \tan^{-1}x + \tan^{-1}y = L.H.S.$$

proof

III. ii) $\cos^{-1}x - \cos^{-1}y = \cos^{-1}(xy + \sqrt{1-x^2}\cdot\sqrt{1-y^2})$

→ let, $\cos^{-1}x = A$ $\qquad$ $\cos^{-1}y = B$

$\cos A = x$ $\qquad$ $\cos B = y$

$L.H.S = \cos^{-1}x - \cos^{-1}y = A - B$

$R.H.S = \cos^{-1}(xy + \sqrt{1-x^2}\cdot\sqrt{1-y^2})$

$= \cos^{-1}(\cos A\cdot\cos B + \sqrt{1-\cos^2 A}\cdot\sqrt{1-\cos^2 B})$

$= \cos^{-1}(\cos A\cdot\cos B + \sin A\cdot\sin B)$

$= \cos^{-1}(\cos(A-B))$

$= A - B$

$\therefore L.H.S = R.H.S$

Date
Page

d) $\sin\left[\sin^{-1}\frac{4}{5} + \cot^{-1}3\right]$

→ Soln,

let $\sin^{-1}\frac{4}{5} = \alpha$

$\sin\alpha = \frac{4}{5} = \frac{p}{h}$

$5 = \sqrt{4^2+b^2}$

$25 = 16 + b^2$

$b^2 = 9$

$b = 3$

$\cos\alpha = \frac{b}{h} = \frac{3}{5}$

let $\cot^{-\cancel{1}} 3 = \beta$

$\cot\beta = \frac{3}{1} = \frac{b}{p}$

$h = \sqrt{b^2+p^2} = \sqrt{1^2+3^2} = \sqrt{10}$

$\sin\beta = \frac{p}{h} = \frac{1}{\sqrt{10}}$, $\cos\beta = \frac{b}{h} = \frac{3}{\sqrt{10}}$

$\sin\left[\sin^{-1}\frac{4}{5} + \cot^{-1}3\right]$

$\sin(\alpha+\beta) = \sin\alpha.\cos\beta + \cos\alpha.\sin\beta$

$= \frac{4}{5}\times\frac{3}{\sqrt{10}} + \frac{3}{5}\times\frac{1}{\sqrt{10}}$

$= \frac{12}{5\sqrt{10}} + \frac{3}{5\sqrt{10}}$

$= \frac{\cancel{15}\,3}{\cancel{5}\sqrt{10}}$

$= \frac{3}{\sqrt{10}}$ Ans.

Date
Page

e) $\tan\left[\cos^{-1}\frac{4}{5} + \tan^{-1}\frac{2}{3}\right]$

→ Soln,

let $\cos^{-1}\frac{4}{5} = A$

$\cos A = \frac{4}{5} = \frac{b}{h}$

$h^2 = p^2 + b^2$

$5^2 = p^2 + 4^2$

$p^2 = 25 - 16$

$p^2 = 9$

$p = 3$

$\sin A = \frac{p}{h} = \frac{3}{5}$

$\tan A = \frac{p}{b} = \frac{3}{4}$

let $\tan^{-1}\frac{2}{3} = B$

$\tan B = \frac{2}{3} = \frac{p}{b}$ → $h = \sqrt{p^2+b^2}$

$h = \sqrt{2^2+3^2}$

$h = \sqrt{4+9}$

$h = \sqrt{13}$

$\sin\beta = \frac{p}{h} = \frac{2}{\sqrt{13}}$, $\cos\beta = \frac{b}{h} = \frac{3}{\sqrt{13}}$

Now,

$\tan\left[\cos^{-1}\frac{4}{5} + \tan^{-1}\frac{2}{3}\right]$

$= \tan(A+B)$

$= \frac{\tan A + \tan B}{1 - \tan A \cdot \tan B}$

$= \frac{\frac{3}{4} + \frac{2}{3}}{1 - \frac{3}{4} \times \frac{2}{3}}$

$$= \frac{\frac{9+8}{\cancel{12}}}{\frac{12-6}{\cancel{12}}}$$

$$= \frac{17}{6}$$

(4) Prove that:

a) $\tan^{-1} 2 - \tan^{-1} 1 = \tan^{-1}\frac{1}{3}$

↳ L.H.S,

$$\tan^{-1} 2 - \tan^{-1} 1$$

$$= \tan^{-1}\left(\frac{2-1}{1+2\times 1}\right)$$

$$= \tan^{-1}\left(\frac{1}{1+2}\right)$$

$$= \tan^{-1}\left(\frac{1}{3}\right) = R.H.S \quad \text{proved}$$

b) $\tan^{-1}\frac{1}{3} + \tan^{-1}\frac{1}{5} + \tan^{-1}\frac{1}{7} + \tan^{-1}\frac{1}{8} = \frac{\pi}{4}$

↳ L.H.S,

$$\tan^{-1}\frac{1}{3} + \tan^{-1}\frac{1}{5} + \tan^{-1}\frac{1}{7} + \tan^{-1}\frac{1}{8}$$

$$= \tan^{-1}\left(\frac{\frac{1}{3}+\frac{1}{5}}{1-\frac{1}{3}\times\frac{1}{5}}\right) + \tan^{-1}\left(\frac{\frac{1}{7}+\frac{1}{8}}{1-\frac{1}{7}\times\frac{1}{8}}\right)$$

$$= \tan^{-1}\left(\frac{\frac{8}{\cancel{15}}}{\frac{14}{\cancel{15}}}\right) + \tan^{-1}\left(\frac{\frac{15}{5\cancel{6}}}{\frac{55}{5\cancel{6}}}\right)$$

$$= \tan^{-1}\left(\frac{8}{14}\right) + \tan^{-1}\left(\frac{15}{55}\right)$$

$$= \tan^{-1}\left(\frac{\frac{8}{14} + \frac{15}{55}}{1 - \frac{8}{14} \times \frac{15}{55}}\right)$$

$$= \tan^{-1}\left(\frac{\frac{65}{77}}{\frac{65\cancel{0}}{77\cancel{0}}}\right)$$

$$= \tan^{-1}\left(\frac{65}{65}\right)$$

$$= \tan^{-1}(1)$$

$$= \tan^{-1}\left(\tan\frac{\pi}{4}\right)$$

$$= \frac{\pi}{4} = \text{R.H.S}$$

$$c)\ \tan^{-1}\frac{m}{n} - \tan^{-1}\left(\frac{m-n}{m+n}\right) = \frac{\pi}{4}$$

= L.H.S,

$$\tan^{-1}\frac{m}{n} - \tan^{-1}\left(\frac{m-n}{m+n}\right)$$

$$= \tan^{-1}\left[\frac{\frac{m}{n} - \frac{m-n}{m+n}}{1 + \frac{m}{n}\times\frac{m-n}{m+n}}\right]$$

$$= \tan^{-1}\left[\frac{\frac{m(m+n) - n(m-n)}{\cancel{n(m+n)}}}{\frac{n(m+n) + m(m-n)}{\cancel{n(m+n)}}}\right]$$

$$= \tan^{-1}\left[\frac{m^2 + \cancel{mn} - \cancel{mn} + n^2}{\cancel{mn} + n^2 + m^2 - \cancel{mn}}\right]$$

$$= \tan^{-1}\left(\frac{m^2+n^2}{m^2+n^2}\right)$$

$$= \tan^{-1}(1)$$

$$= \tan^{-1}\left(\tan\frac{\pi}{4}\right)$$

$$= \frac{\pi}{4} = \text{R.H.S}$$

d) $\tan^{-1}1 + \tan^{-1}2 + \tan^{-1}3 = \pi = 2\left(\tan^{-1}1 + \tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{3}\right)$

→ L.H.S,

$$\tan^{-1}1 + \tan^{-1}2 + \tan^{-1}3$$

$$= \tan^{-1}1 + \tan^{-1}3 + \tan^{-1}2$$

$$= \tan^{-1}\left(\frac{1+3}{1-1\times3}\right) + \tan^{-1}2$$

$$= \tan^{-1}\left(\frac{4}{-2}\right) + \tan^{-1}2$$

$$= \tan^{-1}(-2) + \tan^{-1}2$$

$$= \tan^{-1}\left[\frac{-2+2}{1-2\times-2}\right]$$

$$= \tan^{-1}\left[\frac{0}{1+4}\right]$$

$$= \tan^{-1}(0)$$

$$= \tan^{-1}(\pi)$$

$$= \pi$$

R.H.S.

$$= 2\left(\tan^{-1}1 + \tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{3}\right)$$

$$= 2\left[\tan^{-1}1 + \tan^{-1}\left(\frac{\frac{1}{2}+\frac{1}{3}}{1-\frac{1}{2}\times\frac{1}{3}}\right)\right]$$

$$= 2\left[\tan^{-1}1 + \tan^{-1}\left(\frac{\frac{5}{6}}{\frac{5}{6}}\right)\right]$$

$= 2\left[\tan^{-1} 1 + \tan^{-1} 1\right]$

$= 2\tan^{-1}\left(\frac{1+1}{1-1\times 1}\right)$

$= 2\times\tan^{-1}\left(\frac{2}{0}\right)$

$= 2\times \tan^{-1} \cancel{\frac{\pi}{2}}\ \tan^{-1}\frac{\pi}{2}$

$= \cancel{2}\times\frac{\pi}{\cancel{2}}$

$= \pi$

$\therefore \tan^{-1}1 + \tan^{-1}2 + \tan^{-1}3 = \pi = 2\left(\tan^{-1}1 + \tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{3}\right)$

f) $\tan^{-1}\frac{x-y}{1+xy} + \tan^{-1}\frac{y-z}{1+yz} + \tan^{-1}\frac{z-x}{1+zx} = 0$

$= \cancel{\tan^{-1}x} - \cancel{\tan^{-1}y} + \cancel{\tan^{-1}y} - \cancel{\tan^{-1}z} + \cancel{\tan^{-1}z}$
$- \cancel{\tan^{-1}x}$

$= 0 = R.H.S$

3.c) $\sin^{-1}x - \cos^{-1}(-x)$

Soln,

let $\sin^{-1}x = \theta$.

$\sin\theta = x$

$-\cos\left(\frac{\pi}{2}+\theta\right) = x$.

$\cancel{\cos}\ + \cos\left(\frac{\pi}{2}+\theta\right) = -x$.

$\cos\left(\frac{\pi}{2}+\theta\right) = -x$

$\cos^{-1}(-x) = \frac{\pi}{2} + \theta$

$\therefore \cos^{-1}(-x) = \frac{\pi}{2} + \theta$

Now,

$\sin^{-1}x - \cos^{-1}(-x) = \cancel{\theta} - \frac{\pi}{2} - \cancel{\theta}$

$= -\frac{\pi}{2}$ #.

5. Prove that:

a) $\sin^{-1}(3x - 4x^3) = 3\sin^{-1}x$

$\rightarrow$ ~~R.H.S = $3\sin^{-1}x$~~

let $\sin^{-1}x = \theta$

$\sin\theta = x$.

L.H.S $= \sin^{-1}(3x - 4x^3)$

$= \sin^{-1}(3\sin\theta - 4\sin^3\theta)$

$= \sin^{-1}(\sin 3\theta)$

$= 3\theta$

$= 3\sin^{-1}x$ = R.H.S

Proved

b) $\cos^{-1}(4x^3 - 3x) = 3\cos^{-1}x$.

$\rightarrow$ let $\cos^{-1}x = \theta$

$\cos\theta = x$

L.H.S $= \cos^{-1}(4x^3 - 3x)$

$= \cos^{-1}(4\cos^3\theta - 3\cos\theta)$

$= \cos^{-1}(\cos 3\theta)$

$= 3\theta$

$= 3\cos^{-1}x = R.H.S$

proved

c) $\sec^{-1}x + \operatorname{cosec}^{-1}x = \frac{\pi}{2}$

→ let $\sec^{-1}x = \theta$

$\sec\theta = x.$

$\operatorname{cosec}\left(\frac{\pi}{2} - \theta\right) = x.$

$\operatorname{cosec}^{-1}x = \frac{\pi}{2} - \theta$

$\Rightarrow \operatorname{cosec}^{-1}x + \theta = \frac{\pi}{2}$

$\operatorname{cosec}^{-1}x + \sec^{-1}x = \frac{\pi}{2}$

$\therefore \sec^{-1}x + \operatorname{cosec}^{-1}x = \frac{\pi}{2}$

d) $\tan^{-1}x = \frac{1}{2}\sin^{-1}\frac{2x}{1+x^2}$

→ let $\tan^{-1}x = \theta$

$\tan\theta = x$

$R.H.S = \frac{1}{2}\sin^{-1}\frac{2x}{1+x^2}$

$= \frac{1}{2}\sin^{-1}\frac{2\tan\theta}{1+\tan^2\theta}$

$= \frac{1}{2}\sin^{-1}\sin 2\theta$

$= \frac{1}{\cancel{2}} \times \cancel{2}\theta$

$= \tan^{-1}x = L.H.S$

Date ____
Page ____

e) $2\tan^{-1}x = \sin^{-1}\frac{2x}{1+x^2} = \cos^{-1}\frac{1-x^2}{1+x^2} = \tan^{-1}\frac{2x}{1-x^2}$

→ Soln,

Let $\tan^{-1}x = \theta$

$\tan\theta = x$.

Now,

$\sin^{-1}\frac{2x}{1+x^2} = \sin^{-1}\frac{2\tan\theta}{1+\tan^2\theta}$

$= \sin^{-1}\sin 2\theta$

$= 2\theta$

$= 2\tan^{-1}x$

$\cos^{-1}\frac{1-x^2}{1+x^2} = \cos^{-1}\frac{1-\tan^2\theta}{1+\tan^2\theta}$

$= \cos^{-1}\cos 2\theta$

$= 2\theta$

$= 2\tan^{-1}x$

$\tan^{-1}\frac{2x}{1-x^2} = \tan^{-1}\frac{2\tan\theta}{1-\tan^2\theta}$

$= \tan^{-1}\tan 2\theta$

$= 2\theta$

$= 2\tan^{-1}x$ #.

f) $\tan(2\tan^{-1}x) = 2\tan(\tan^{-1}x + \tan^{-1}x^3) = \frac{2x}{1-x^2}$

→ Soln,

let $\tan^{-1}x = \theta$

$\tan\theta = x$

Now,

$$\frac{2x}{1-x^2} = \frac{2\tan\theta}{1-\tan^2 x}$$

$$= \tan 2\theta$$

$$= \tan(2\tan^{-1}x)$$

$$2\tan(\tan^{-1}x + \tan^{-1}x^3)$$

$$= 2\tan\left(\tan^{-1}\left(\frac{x+x^3}{1-x\times x^3}\right)\right)$$

$$= 2\tan\left(\tan^{-1}\left(\frac{x(1+x^2)}{1-x^4}\right)\right)$$

$$= \frac{2x\cancel{(1+x^2)}}{(1-x^2)\cancel{(1+x^2)}}$$

$$= \frac{2x}{1-x^2}$$

Date ______
Page ______

2.c) $\sin^{-1}\sin\frac{2\pi}{3}$

$= \sin^{-1}\sin\left(\pi-\frac{\pi}{3}\right) \quad \therefore y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$= \sin^{-1}\sin\frac{\pi}{3}$

$= \frac{\pi}{3}$

5.g) $\sin^{-1}\sqrt{\frac{x-b}{a-b}} = \cos^{-1}\sqrt{\frac{a-x}{a-b}} = \tan^{-1}\sqrt{\frac{x-b}{a-x}}$

→ Soln,

Let $\sin^{-1}\sqrt{\frac{x-b}{a-b}} = A$

$\sin A = \sqrt{\frac{x-b}{a-b}}$

$\sin^2 A = \frac{x-b}{a-b}$

$1-\cos^2 A = \frac{x-b}{a-b}$

$\cos^2 A = 1 - \frac{x-b}{a-b}$

$\cos^2 A = \frac{a-\cancel{b}-x+\cancel{b}}{a-b}$

$\cos^2 A = \frac{a-x}{a-b}$

$\cos A = \sqrt{\frac{a-x}{a-b}} \Rightarrow \cos^{-1}\sqrt{\frac{a-x}{a-b}} = A$

Now,

$\tan^2 A = \frac{\sin^2 A}{\cos^2 A}$

$$\tan^2 A = \frac{\dfrac{x-b}{\cancel{a-b}}}{\dfrac{a-x}{\cancel{a-b}}}$$

$$= \frac{x-b}{a-x}$$

$$\tan A = \sqrt{\frac{x-b}{a-x}}$$

$$\tan^{-1}\sqrt{\frac{x-b}{a-x}} = A$$

$$\therefore \sin^{-1}\sqrt{\frac{x-b}{a-b}} = \cos^{-1}\sqrt{\frac{a-x}{a-b}} = \tan^{-1}\sqrt{\frac{x-b}{a-x}}$$

Date
Page

i) $\cot^{-1}x + \cot^{-1}y = \cot^{-1}\dfrac{xy-1}{y+x}$

ii) $\cot^{-1}x - \cot^{-1}y = \cot^{-1}\dfrac{xy+1}{y-x}$

iii) $\tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \tan^{-1}\dfrac{x+y+z-xyz}{1-xy-yz-zx}$

	Inverse Circular Function	Domain	Range [Principal Value]
i)	$y=\sin^{-1}x$	$[-1,1]$	$\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$
ii)	$y=\cos^{-1}x$	$[-1,1]$	$[0,\pi]$
iii)	$y=\tan^{-1}x$	$(-\infty,\infty)$	$\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$
iv)	$y=\cot^{-1}x$	$(-\infty,\infty)$	$(0,\pi)$
v)	$y=\sec^{-1}x$	$\lvert x\rvert\geq 1$	$[0,\pi],\ y\neq\frac{\pi}{2}$
vi)	$y=\operatorname{cosec}^{-1}x$	$\lvert x\rvert\geq 1$	$\left[-\frac{\pi}{2},\frac{\pi}{2}\right],\ y\neq 0$

ii) Proof

let $\cot^{-1}x = A$
$\cot A = x$

let $\cot^{-1}y = B$
$\cot B = y$

$L.H.S = \cot^{-1}x - \cot^{-1}y$
$= A - B$

Date
Page

$$\text{R.H.S} = \cot^{-1}\left(\frac{xy+1}{y-x}\right)$$

$$= \cot^{-1}\left(\frac{\cot A \cdot \cot B + 1}{\cot B - \cot A}\right)$$

$$= \cot^{-1}(\cot(A-B))$$

$$= A-B$$

8

a) If $\tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \pi$, show that $x+y+z = xyz$.

↳ Soln,

$$\tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \pi$$

or, $\tan^{-1}\left(\frac{x+y}{1-xy}\right) + \tan^{-1}z = \pi$

or, $\tan^{-1}\left(\frac{\frac{x+y}{1-xy} + z}{1 - \left(\frac{x+y}{1-xy}\right)z}\right) = \pi$

or, $\tan^{-1}\left(\frac{\frac{x+y+z-xyz}{\cancel{1-xy}}}{\frac{1-xy-xz-yz}{\cancel{1-xy}}}\right) = \pi$

or, $\tan^{-1}\left(\frac{x+y+z-xyz}{1-xy-xz-yz}\right) = \tan^{-1}0$

or, $\frac{x+y+z-xyz}{1-xy-xz-yz} = 0$

or, $x+y+z-xyz = 0$

$\therefore x+y+z = xyz$

Proved

b) If $\cot^{-1}x + \cot^{-1}y + \cot^{-1}z = \pi$, show that $xy + yz + zx = 1$.

Soln,

$$\cot^{-1}x + \cot^{-1}y + \cot^{-1}z = \pi$$

or, $\tan^{-1}\frac{1}{x} + \tan^{-1}\frac{1}{y} + \tan^{-1}\frac{1}{z} = \pi$

or, $\tan^{-1}\left(\frac{\frac{1}{x}+\frac{1}{y}}{1-\frac{1}{x}\times\frac{1}{y}}\right) + \tan^{-1}\frac{1}{z} = \pi$

or, $\tan^{-1}\left(\frac{\frac{x+y}{\cancel{xy}}}{\frac{xy-1}{\cancel{xy}}}\right) + \tan^{-1}\frac{1}{z} = \pi$

or, $\tan^{-1}\left(\frac{\frac{x+y}{xy-1}+\frac{1}{z}}{1-\left(\frac{x+y}{xy-1}\right)\times\frac{1}{z}}\right) = \pi$

or, $\tan^{-1}\left(\frac{\frac{xz+yz+xy-1}{\cancel{z(xy-1)}}}{\frac{xyz-z-x-y}{\cancel{z(xy-1)}}}\right) = \pi$

or, $\tan\pi = \frac{xz+yz+xy-1}{xyz-x-z-y}$

or, $0 = xz+yz+xy-1$

$\therefore xz + xy + yz = 1$

Proved

Date
Page

6.

a) $\cos^{-1} x - \sin^{-1} x = 0$

or, $\cos^{-1} x = \sin^{-1} x$

or, $\cos^{-1} x = \cos^{-1} \sqrt{1-x^2}$

or, $x = \sqrt{1-x^2}$

or, $x^2 = 1 - x^2$

or, $2x^2 = 1$

or, $x^2 = \frac{1}{2}$

or, $x = \pm \frac{1}{\sqrt{2}}$

$\therefore x = \frac{1}{\sqrt{2}}$ $\left(\therefore x = \frac{-1}{\sqrt{2}} \text{ is not possible}\right)$.

b) $\sin^{-1} \frac{1}{2} x = \cos^{-1} x$

or, $\sin^{-1} \frac{1}{2} x = \sin^{-1} \sqrt{1-x^2}$

or, $\frac{x}{2} = \sqrt{1-x^2}$

or, $\frac{x^2}{4} = 1 - x^2$

or, $x^2 = 4 - 4x^2$

or, $4x^2 + x^2 = 4$

or, $5x^2 = 4$

or, $x^2 = \frac{4}{5}$

$\therefore x = \frac{2}{\sqrt{5}}$ $\left(\therefore x = \frac{-2}{\sqrt{5}} \text{ is not possible}\right)$.

c) $\cos^{-1} x = \cos^{-1} \frac{1}{2x}$

or, $x = \frac{1}{2x}$

or, $2x^2 = 1$

or, $x^2 = \frac{1}{2}$

$x = \sqrt{\frac{1}{2}}$

$\therefore x = \pm \frac{1}{\sqrt{2}}$

7.

a) If $\sec^{-1} x = \operatorname{cosec}^{-1} y$, prove that $\frac{1}{x^2} + \frac{1}{y^2} = 1$

→ Soln,

$\sec^{-1} x = \operatorname{cosec}^{-1} y$

~~sec x~~

$\cos^{-1} \frac{1}{x} = \sin^{-1} \frac{1}{y}$

$\cos^{-1} \frac{1}{x} = \cos^{-1} \sqrt{1 - \frac{1}{y^2}}$

$\frac{1}{x} = \sqrt{1 - \frac{1}{y^2}}$

$\frac{1}{x^2} = 1 - \frac{1}{y^2}$

$\frac{1}{x^2} + \frac{1}{y^2} = 1$

pr

Date
Page

d) $\tan^{-1}x - \cot^{-1}x = 0$.

or, $\tan^{-1}x - \tan^{-1}\frac{1}{x} = 0$

or, $\tan^{-1}\dfrac{x - \frac{1}{x}}{1 + \cancel{x} \times \frac{1}{\cancel{x}}} = 0$

or, $\tan^{-1}\dfrac{\frac{x^2-1}{x}}{2} = 0$

or, $\tan^{-1}\dfrac{x^2-1}{2x} = 0$

or, $\tan 0 = \dfrac{x^2-1}{2x}$

or, $0 = \dfrac{x^2-1}{2x}$

$x^2 - 1 = 0$

$x^2 = 1$

$x = \sqrt{1}$

$x = \pm 1$

$\therefore x = 1$ [$\because -1$ is not possible]

e) $\sin^{-1}2x - \sin^{-1}\sqrt{3}x = \sin^{-1}x$ $(x>0)$

$\Rightarrow \sin^{-1}2x - \sin^{-1}x = \sin^{-1}\sqrt{3}x$

or, $\sin^{-1}(2x\sqrt{1-x^2} + x\sqrt{1-4x^2} = \sin^{-1}\sqrt{3}x$

or, $\cancel{\sin^{-1}}(2x\sqrt{1-x^2} + x\sqrt{1-4x^2} = \sqrt{3}x$

or, $\cancel{x}[2\sqrt{1-x^2} + \sqrt{1-4x^2}] = \sqrt{3}\cancel{x}$

or, $2\sqrt{1-x^2} + \sqrt{1-4x^2} = \sqrt{3}$ [$\because x>0$]

or, $2\sqrt{1-x^2} - \sqrt{3} = -\sqrt{1-4x^2}$

Squaring on both sides,

or, $4(1-x^2) - 2\cdot 2\sqrt{1-x^2} \times \sqrt{3} + 3 = 1-4x^2$

or, $4-\cancel{4x^2} - 4\sqrt{3-3x^2} + 3 = 1-\cancel{4x^2}$

or, $\cancel{7} \quad 7 - 4\sqrt{3-3x^2} = 1$

or, $6 - 4\sqrt{3-3x^2} = 0$

or, $3 - 2\sqrt{3-3x^2} = 0$

or, $3 = 2\sqrt{3-3x^2}$

Squaring on both side,

$9 = 4(3-3x^2)$

$9 = 12 - 12x^2$

or, $12x^2 = 12-9$

or, $12x^2 = 3$

or, $x^2 = \frac{\cancel{3}}{\cancel{12}\,4}$

or, $x = \sqrt{\frac{1}{4}}$

$\therefore x = \frac{1}{2}$ $\quad$ $[\because x = -\frac{1}{2}$ is not possible$]$.

f) $\sin^{-1}\frac{2a}{1+a^2} - \cos^{-1}\frac{1-b^2}{1+b^2} = 2\tan^{-1}x$.

let $a = \tan\alpha$ and $b = \tan\beta$

Now,

$L.H.S = \sin^{-1}\frac{2a}{1+a^2} - \cos^{-1}\frac{1-b^2}{1+b^2}$

$= \sin^{-1}\frac{2\tan\alpha}{1+\tan^2\alpha} - \cos^{-1}\frac{1-\tan^2\beta}{1+\tan^2\beta}$

$= \sin^{-1}\sin 2\alpha - \cos^{-1}\cos 2\beta$

$= 2\alpha - 2\beta$

$= 2(\alpha - \beta)$

Date
Page

$= 2(\tan^{-1} a - \tan^{-1} b)$

$= 2 \tan^{-1}\left(\frac{a-b}{1+ab}\right)$

$\therefore 2\tan^{-1}\left(\frac{a-b}{1+ab}\right) = 2\tan^{-1} x$

or, $\tan^{-1}\left(\frac{a-b}{1+ab}\right) = \tan^{-1} x$

$\therefore x = \frac{a-b}{1+ab}$ #.

Date
Page

9.

a) If $\sin^{-1}x + \sin^{-1}y + \sin^{-1}z = \frac{\pi}{2}$, prove that $x^2+y^2+z^2+2xyz=1$.

→ Given,

$$\sin^{-1}x + \sin^{-1}y + \sin^{-1}z = \frac{\pi}{2}$$

Let, $\sin^{-1}x = A \Rightarrow x = \sin A$

$\sin^{-1}y = B \Rightarrow y = \sin B$

$\sin^{-1}z = C \Rightarrow z = \sin C$

Then,

$$A+B+C = \frac{\pi}{2}$$

$$A+B = \frac{\pi}{2} - C$$

$$\cos(A+B) = \cos\left(\frac{\pi}{2} - C\right)$$

or, $\cos A \cdot \cos B - \sin A \cdot \sin B = \sin C$

or, $\sqrt{1-\sin^2 A} \cdot \sqrt{1-\sin^2 B} - xy = z$

or, $\sqrt{1-x^2} \cdot \sqrt{1-y^2} - xy = z$

or, $\sqrt{1-x^2-y^2+x^2y^2} = z + xy$

Squaring on both sides,

$$1-x^2-y^2+\cancel{x^2y^2} = z^2+2xyz+\cancel{x^2y^2}$$

or, $x^2+y^2+z^2+2xyz=1$

$\therefore x^2+y^2+z^2+2xyz=1$

Proved.

b) If $\sin^{-1}x + \sin^{-1}y + \sin^{-1}z = \pi$, prove that $x\sqrt{1-x^2} + y\sqrt{1-y^2} + z\sqrt{1-z^2} = 2xyz$.

→ Given,

$\sin^{-1}x = A \Rightarrow x = \sin A$

$\sin^{-1}y = B \Rightarrow y = \sin B$

$\sin^{-1}z = C \Rightarrow z = \sin C$.

$\sin^{-1}x + \sin^{-1}y + \sin^{-1}z = \pi$

$A + B + C = \pi \Rightarrow A + B = \pi - C$

or, ~~$A+B = \pi - C$~~

or, ~~$\sin(A+B) = \sin(\pi - C)$~~

or, ~~$\sin A \cdot \cos B + \cos A \cdot \sin B =$~~

L.H.S.

$= x\sqrt{1-x^2} + y\sqrt{1-y^2} + z\sqrt{1-z^2}$

$= \sin A\sqrt{1-\sin^2 A} + \sin B\sqrt{1-\sin^2 B} + \sin C\sqrt{1-\sin^2 C}$

$= \sin A \cdot \cos A + \sin B \cdot \cos B + \sin C \cdot \cos C$.

$= \frac{1}{2} \times 2\sin A \cdot \cos A + \frac{1}{2} \times 2\sin B \cdot \cos B + \frac{1}{2} \times 2\sin C \cdot \cos C$

$= \frac{1}{2}\sin 2A + \frac{1}{2}\sin 2B + \frac{1}{2}\sin 2C$

$= \frac{1}{2}[\sin 2A + \sin 2B + \sin 2C]$

$= \frac{1}{2}\left[2\sin\left(\frac{2A+2B}{2}\right) \cdot \cos\left(\frac{2A-2B}{2}\right) + \sin 2C\right]$

$= \frac{1}{2}\left[2\sin(A+B) \cdot \cos(A-B) + \sin 2C\right]$

$= \frac{1}{\cancel{2}}[\cancel{2}\sin(A+B) \cdot \cos(A-B)] + \frac{1}{\cancel{2}} \times \cancel{2}\sin C \cdot \cos C$

$= [\sin(\pi - C) \cdot \cos(A-B)] + \sin C \cdot \cos C$

$= \cancel{\cos} \sin C \cdot \cos(A-B) + \sin C \cdot \cos C$

Date
Page

$= \sin C\,[\cos(A-B) + \cos[\pi-(A+B)]]$

$= \sin C\,[\cos(A-B) - \cos(A+B)]$

$= \sin C\,[2\sin A \cdot \sin B]$

$= 2\sin A \cdot \sin B \cdot \sin C$

$= 2xyz$ $\frac{1}{2}$.

6. g) $\tan^{-1}(x+1) + \tan^{-1}(x-1) = \tan^{-1}\left(\frac{8}{31}\right)$

→ Soln,

$\tan^{-1}\left(\frac{x+\cancel{1}+x-\cancel{1}}{1-(x+1)(x-1)}\right) = \tan^{-1}\left(\frac{\cancel{8}}{31}\right)$

or, $\tan^{-1}\left(\frac{2x}{1-x^2+1}\right) = \tan^{-1}\left(\frac{8}{31}\right)$

or, $\frac{\cancel{2}x}{2-x^2} = \frac{\cancel{8}4}{31}$

or, $8-4x^2 = 31x$

or, $4x^2+31x-8=0$

or, $4x^2+32x-x-8=0$

or, $4x(x+8)-1(x+8)=0$

or, $(4x-1)(x+8)=0$

Either,

$x+8=0$

$x=-8$ (Impossible)

$4x-1=0$

$4x=1$

$x=\frac{1}{4}$

$\therefore x=\frac{1}{4}$ 4.

Scanned with CamScanner

Date
Page

7.

b) Prove that: $\sin^{-1}x + \sin^{-1}y = \cos^{-1}\left[\sqrt{(1-x^2)(1-y^2)} - xy\right]$

Soln,

Let $\sin^{-1}x = A$ and $\sin^{-1}y = B$

$\sin A = x$ $\quad\quad$ $\sin B = y$

$$R.H.S = \cos^{-1}\left[\sqrt{(1-x^2)(1-y^2)} - xy\right]$$
$$= \cos^{-1}\left[\sqrt{(1-\sin^2A)(1-\sin^2B)} - \sin A\cdot\sin B\right]$$
$$= \cos^{-1}\left[\sqrt{\cos^2A\cdot\cos^2B} - \sin A\cdot\sin B\right]$$
$$= \cos^{-1}\left[\cos A\cdot\cos B - \sin A\cdot\sin B\right]$$
$$= \cos^{-1}\left[\cos(A+B)\right]$$
$$= A+B$$
$$= \sin^{-1}x + \sin^{-1}y = L.H.S$$

c) Prove that: $\sin^{-1}x + \cos^{-1}y = \sin^{-1}\left[xy + \sqrt{(1-x^2)(1-y^2)}\right]$

Soln,

Let $\sin^{-1}x = A$ And $\cos^{-1}y = B$

$\sin A = x$ $\quad\quad$ $\cos B = y$

$$R.H.S = \sin^{-1}\left[xy + \sqrt{(1-x^2)(1-y^2)}\right]$$
$$= \sin^{-1}\left[\sin A\cdot\cos B + \sqrt{(1-\sin^2A)(1-\cos^2B)}\right]$$
$$= \sin^{-1}\left[\sin A\cdot\cos B + \sqrt{\cos^2A\cdot\sin^2B}\right]$$
$$= \sin^{-1}\left[\sin A\cdot\cos B + \cos A\cdot\sin B\right]$$
$$= \sin^{-1}\left[\sin(A+B)\right]$$
$$= A+B$$
$$= \sin^{-1}x + \cos^{-1}y = L.H.S$$

Proved

Date
Page

d) Prove that: $\cot^{-1}x - \cot^{-1}y = \cot^{-1}\frac{xy+1}{y-x}$

↳ Let $\cot^{-1}x = A$ And $\cot^{-1}y = B$

$\cot A = x$ $\cot B = y$

$$R.H.S = \cot^{-1}\left(\frac{xy+1}{y-x}\right)$$

$$= \cot^{-1}\left(\frac{\cot A \cdot \cot B + 1}{\cot B - \cot A}\right)$$

$$= \cot^{-1}(\cot(A-B))$$

$$= A - B$$

$$= \cot^{-1}x - \cot^{-1}y = L.H.S$$

Proved.

⊛ MCQ

1) The solution of the equation $\cos^{-1}(x\sqrt{3}) + \cos^{-1}x = \frac{\pi}{2}$ is given by

Ans: c) $x = \frac{1}{2}$

2) If $4\sin^{-1}x + \cos^{-1}x = \pi$, the value of x is equal to

Ans: b) $\frac{1}{2}$

3) If $\tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \pi$ then $x+y+z =$

Ans: a) xyz.

Date
Page

4) If $\sin^{-1}x = \frac{\pi}{3}$ then $\cos^{-1}x =$

Ans b) $\frac{\pi}{6}$

5) If $\sin^{-1}x + \sin^{-1}y = \frac{2\pi}{3}$ then $\cos^{-1}x + \cos^{-1}y =$

Ans c) $\pi/3$

6) If $x = \sin^{-1}k$, $y = \cos^{-1}k$, $-1 \le k \le 1$, then the correct relationship is

Ans c) $x + y = \frac{\pi}{2}$

7) If $\tan(x+y) = 33$ and $x = \tan^{-1}3$, then y is equal to.

Ans b) $\tan^{-1}\frac{3}{10}$.

8) If $\cos(2\sin^{-1}x) = \frac{1}{9}$ then $x =$

Ans c) $\pm\frac{2}{3}$

9) $\tan^{-1}\left(\frac{a-b}{1+ab}\right) + \tan^{-1}\left(\frac{b-c}{1+bc}\right)$ is equal to

Ans c) $\tan^{-1}a - \tan^{-1}c$

10) The value of $\sec^{-1}\left(\frac{2}{\sqrt{3}}\right)$ is

Ans: c) $\frac{\pi}{6}$

11) The value of $\sin\left(\cos^{-1}\frac{3}{5}\right)$ is

Ans: b) $\frac{4}{5}$

12) The principal value of $\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)$ is

Ans: b) $-\frac{\pi}{3}$